Deep Learning-Based Detection of Stroke-Induced Dysarthria Using Audio Spectrogram Representations

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***Abstract*—Dysarthria is a motor speech disorder that often results from neurological damage, such as a stroke. Clinical diagnostics could greatly benefit from automated speech signal detection of stroke-induced dysarthria. In order to distinguish stroke patients from healthy controls, this research provides a deep learning-based approach that uses 2D spectrograms of speech utterances. We assessed various phonetic difficulties, such as individual vowels, bisyllabic phrases, and entire sentences, using a clinically recorded dataset from the Indian Council of Medical Research (ICMR). We transformed 1D audio signals into Mel-spectrograms and employed pre-trained 2D convolutional neural networks (MobileNetV2 and ResNet-50) using a 5-fold cross-validation strategy. The 2D architectures were systematically compared against baseline 1D-MobileNet and custom 1D-CNN models. Our results demonstrate that converting audio into rich 2D time-frequency representations enables standard image-classification networks to achieve robust classification metrics, with MobileNet-2D yielding high sensitivity, especially in vowel classification (up to 96.25%), words (up to 94.46%), and sentences (S0 – 92.5%).**

***Index Terms*—Dysarthria, Stroke Detection, Deep Learning, Spectrograms, MobileNet, ResNet, Speech Processing**

I. Introduction

Stroke is a life-threatening medical condition caused by the sudden rupture or blockage of blood vessels supplying oxygen to the brain, leading to localized brain cell death. Depending on the affected cerebral region, a stroke can result in various motor and non-motor impairments. One of the most prevalent motor impairments is dysarthria, a speech disorder characterized by muscle weakness, slurred speech, and prosodic abnormalities [1], [2].

Dysarthria is a prevalent motor speech disorder characterized by impaired muscle control over the speech mechanism, resulting in reduced speech intelligibility, slurred articulation, and altered prosody [3], [4]. It commonly arises from neurological conditions such as stroke, Parkinson's disease, and traumatic brain injury, affecting millions globally.

The subjective assessments of speech-language pathologists used in traditional clinical assessment of dysarthria are time-consuming, prone to inter-rater variability, and have limited scalability, particularly in settings with low resources. By enabling early intervention, tailored rehabilitation, and remote monitoring, automated identification and classification of dysarthric speech can improve the quality of life for those who are impacted [5], [6]. Dysarthria is a major public health concern in India, where stroke is a major cause of long-term impairment with the estimated prevalence of 1.25–1.8 million cases per year, especially among Telugu-speaking people in Andhra Pradesh and Telangana. Based on population-adjusted incidence rates (e.g., 119–145 per 100,000 people, applied to India's 1.4 billion people), the estimate of 1.5–2 million annual cases seems to be a reasonable approximation. However, recent data from 2021–2024 suggests a slightly lower range of 1.25–1.8 million new cases per year. Due to the high stroke rates in southern regions like Andhra Pradesh and Telangana, regional studies in Telugu-speaking areas have shown that dysarthria affects 30–50% of stroke survivors worldwide and in India. [7]

Traditionally, skilled speech-language pathologists use perceptual examination to diagnose and assess the severity of stroke-induced dysarthria [8]. However, this procedure can be subjective and resource-intensive. Automated speech analysis offers an objective, non-invasive biomarker for stroke identification. Early computational techniques depended on hand-crafted acoustic features. However, current breakthroughs in deep learning have dramatically improved classification frameworks [9].

Early techniques used classical acoustic features like Mel-Frequency Cepstral Coefficients (MFCCs) in conjunction with classifiers like Gaussian Mixture Models (GMMs) or Support Vector Machines [10]. These approaches, however, frequently perform poorly on restricted, noisy clinical data due to their inability to capture complicated temporal and spectral patterns. The use of deep learning (DL) paradigms, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs) [11], and pre-trained models, has significantly improved accuracy by exploiting hierarchical feature learning from raw audio or spectrograms. Spectrograms are capable of

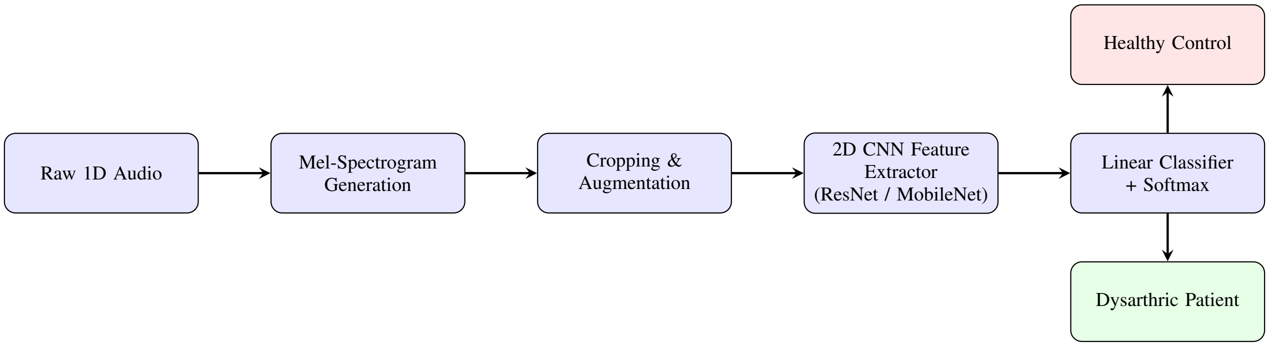

Fig. 1. End-to-End Pipeline for Stroke-Induced Dysarthria Detection using 2D Spectrograms.

being employed to effectively represent audio signals in two dimensions, by capturing both time and frequency domains simultaneously.

This research presents 2D image classification methodologies in which mel-spectrograms generated by clinical speech recordings are fed into ResNet-50 and MobileNetV2 models to uncover discriminative patterns that classify dysarthric speech. We evaluated the proposed 2D classification models on a clinical dataset consisting of sentences, vowels, and words, benchmarking their performance against previously established 1D baselines.

II. Related Work

The availability of corpora, which are frequently limited and biased toward specific etiologies or recording conditions, is a critical component of dysarthric speech research. [12] developed the Universal Access (UA)-Speech corpus to target dysarthric automatic speech recognition (DASR) for cerebral palsy (CP) patients. It contains audio-visual recordings of 255 words (e.g., alphabets, numerals, commands, and Brown corpus words) from 15 CP patients and 4 healthy controls, with intelligibility labeled using human listener scores. While effective for phoneme-level tasks, [13] showed that employing HMM-GMM systems on UA-Speech digits, time-scaling techniques increase word recognition by up to 15%, emphasizing its relevance for augmentation in low-intelligibility settings.

Similarly, the TORGO corpus, contains electromagnetic articulography (EMA) and acoustic data from 7 CP and 1 ALS patients (aged 16-65), who were matched with 8 controls. This data includes sentences from TIMIT and Yorkston-Beukelman intelligibility scoring, words like numbers & commons, and sub-word units like sustained vowels, Consonant-Vowel (CV) syllables [14]. However, the recording settings were critiqued by Schü et al. [15] being it is biased and pointed that classifiers might learn ambient artifacts instead of actual dysarthric characteristics, which limits the cross-corpus generalizability. According to preliminary findings, dysarthric samples had lower speech rates. But rather than stroke victims, they mostly include speech from those with cerebral palsy.

In the Indian context, language diversity and resource constraints make these challenges worsen. Thoppil et al. [16] created a Malayalam dysarthric dataset from 108 patients across six dysarthria types (ataxic, spastic, etc.). Authors have used sustained /a/ phonations and machine learning for pathology categorization and achieved 88% accuracy with SVM classification.

Using k-NN classifiers on spectral characteristics, Jothilakshmi et al. achieved 90%+ accuracy on a Tamil corpus of /a/ vowels from 108 patients and 255 controls. These studies draw attention to the dearth of Dravidian-language materials, which drives the development of databases tailored to certain regions [17].

Recent studies have focused specifically on stroke-induced dysarthria in clinical settings. Banerjee et al. [6] developed a speaker-independent categorization framework, emphasizing the need for reliable models that can manage clinical data variation. Similarly, Javanmardi et al. demonstrated the effectiveness of transfer learning in this field by using pre-trained models to identify and categorize dysarthria from speech [18].

The categorization of neurodegenerative disorders has shown great success with vowel phonation translation into spectrogram images. Compared to conventional machine learning classifiers based on hand-crafted features, the fine-tuning pre-trained CNNs (such as MobileNetV2 and ResNet) may produce higher classification accuracies. Building on these insights, we apply the 2D spectrogram classification paradigm confined to stroke-induced dysarthria, thoroughly evaluating it against 1D CNN and 1D MobileNet methods over variable lengths of utterances.

III. Spectrogram Generation and Augmentation

The raw audio waveforms (1D) were converted into 2D visual representations i.e, spectrograms. The Spectrograms capture the characteristics of time and frequencies of the audio signal, these image representations are fed to 2D deep learning architectures for further training the models.

A. *Pre-processing Pipeline*

The audio samples were processed with default librosa package with default sampling rate of 22.05 kHz. Spectral analysis was carried out with a window size (n_{fft}) of 2048 samples, which corresponds to around 93 milliseconds. A hop length of 512 samples (about 23 ms) was used to achieve enough temporal resolution and overlap between subsequent frames. The top frequency limit was set to 5,000 Hz to focus on the frequency components that are most significant to speech characteristics. The resulting spectrum representations were computed using the Mel scale, and the power spectrograms were subsequently converted to the decibel (dB) domain to reflect the logarithmic nature of human auditory perception. The spectrograms for word /PAPA/ and vowel /a/ for both control and dysarthic speech are shown in figures Fig.2 and Fig.3 respectively. The axes and margins were removed for the spectrograms corresponding to each utterance, and these were used as inputs to the deep learning models.

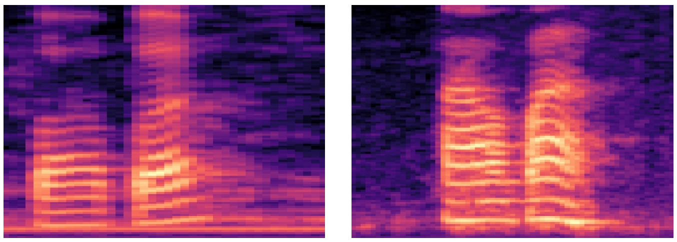

Fig. 2. Sample Mel-spectrograms of control (left) and dysarthric(right) for word PAPA.

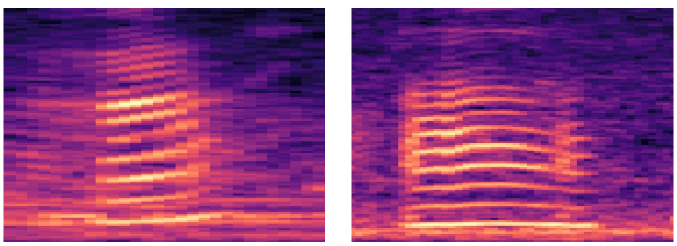

Fig. 3. Sample Mel-spectrograms of control (left) and dysarthric(right) for vowel /a/.

B. Data Augmentation and Cropping

Given the limited size of clinical datasets, data augmentation was applied directly to the generated 2D images to improve model generalization. A horizontal flip (`Image.FLIP_LEFT_RIGHT`) was applied to the spectrograms to create augmented samples, thereby doubling the dataset size for training.

To remove redundant padding and isolate the active speech features, the spectrogram images were cropped. Specifically, 20 pixels were removed from both the left and right edges, and 45 pixels were removed from the top edge, framing the core time-frequency activation patterns before feeding them into the CNN architectures.

IV. Methodology

The core methodology involves the application of transfer learning using established 2D image classification architectures fine-tuned on our specialized spectrogram dataset. The proposed approach block diagram is shown in Fig.1.

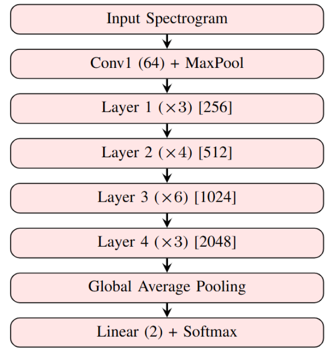

Fig. 4. The ResNet-2D classification pipeline architecture.

A. ResNet-2D Architecture

We used ResNet-50, a deep convolutional neural network that effectively trains deep architectures through residual learning [19]. The model has over 23.5 million trainable parameters and can handle $224 \times 224 \times 3$ input images. The initial convolutional layer is followed by spatial downsampling, max-pooling for early feature extraction, batch normalization, and ReLU activation. The network is divided into four subsequent phases consisting of bottleneck residual blocks to minimize vanishing gradient issues and optimize gradient propagation. 1×1, 3×3, and 1×1 convolutions with identity or projection shortcuts are included in each stage. Spatial resolution decreases as feature dimensionality increases over stages. The design is completed with adaptive average pooling and a dropout-regularized fully connected layer, which enable accurate hierarchical feature extraction from spectrogram representations where particular formant transitions and spectrum deterioration are frequently seen in dysarthric speech.

The ResNet-2D architecture is shown in Fig.4.

B. MobileNet-2D (MobileNetV2)

MobileNetV2 was preferred for deployment in resource-constrained environments, such as mobile health systems, because to its lightweight and computationally efficient design [20]. To achieve reliable high performance with far fewer parameters than traditional convolutional networks, the architecture uses depthwise separable convolutions and inverted residual blocks. MobileNetV2 is used in this study as an effective feature extractor for speech mel-spectrogram representations. Using an initial Conv2D layer (32 channels) and successive bottleneck stages at 24, 32, 96, and 320 channels, the modified 2D pipeline processes input spectrograms of size $224 \times 224 \times 3$. These steps perform spatial downsampling while gradually capturing hierarchical time-frequency information. The network concludes with a Conv2D expansion layer (1280 channels), global average pooling, and a binary classification head, which allows for fast real-time dysarthria detection. The architecture is shown in Fig.5

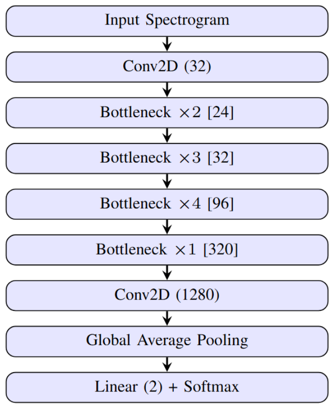

Fig. 5. The MobileNet-2D classification pipeline architecture.

V. Experimental Setup

A. Dataset

In this work, we used the dataset developed by Banerjee et al. [6], a clinically recorded Telugu dysarthric speech corpus collected at AIIMS Mangalagiri, Andhra Pradesh. The study's dataset consists of high-quality audio recordings (44.1 kHz, 16-bit) from 50 stroke patients (aged 30-84, mean 65 years) and 50 healthy controls, all of whom are native Telugu speakers. Recordings were made in real clinical ward situations with an iPad. The corpus is organized into three phonetic categories:
Vowels include five sustained phonations (/a/, /e/, /i/, /o/, and /u/), each repeated twice.
Words: Three two-syllable plosive utterances (PAPA, PIPI, and PUPU), each repeated twice.
Sentences: Ten brief phonetically rich Telugu sentences (maximum of three words each; S01 and S02 are typical among speakers).

Clinicians labeled the degree of dysarthria and used NIHSS scores to gauge the severity of the stroke. This dataset supports both text-dependent and speaker-independent classification studies.

B. Training and Evaluation Strategy

The dataset was separated into five discrete folds for k-fold cross-validation. The training and test speakers are independent for each split. Training was conducted across 30 epochs in batches of 20. During DataLoader instantiation, a `StratifiedSampler` was employed to handle any class imbalances between stroke patients and healthy controls.

A learning rate of 1×10^{-5} was set for the models. Classification accuracy (%), specificity (%), and sensitivity (%) were used to evaluate the model's performance.

VI. Results

According to the experimental findings, baseline 1D methods such a custom CNN and MobileNet-1D are on par with the proposed 2D models, ResNet-2D and MobileNet-2D. Performance was evaluated on speech tasks that were categorized as vowels, words, and sentences that were varying in phonetic difficulty.

A. Performance on Vowels

The performance for each of the five sustained vowels is shown in Table I. With a sensitivity of 96.25% for the vowel "A" and 92.5% for the vowel "O," MobileNet-2D proved to be highly robust in detecting stroke patients (see Figure 6). ResNet-2D achieved the maximum accuracy for 'E' (89.00%), while the custom 1D-CNN and ResNet-2D offered an optimal balance. Notably, MobileNet-1D achieved high specificity (up to 100% for 'E') but lagged in sensitivity compared to the 2D models.

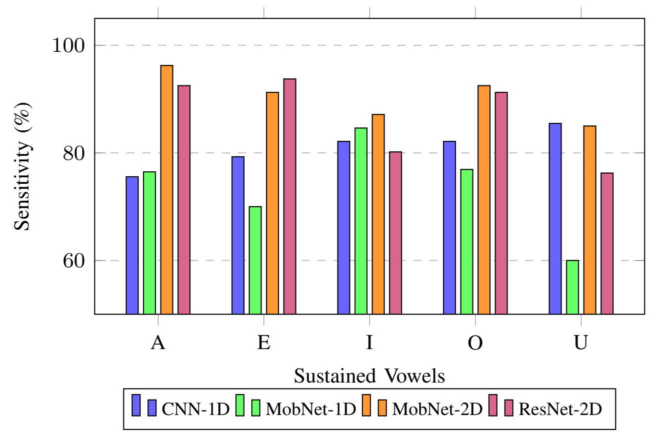

Fig. 6. Comparison of Model Sensitivity across isolated vowel phonations.

B. Performance on Bi-syllabic Words

When analyzing bi-syllabic words (PAPA, PIPI, PUPU) in Table II, MobileNet-2D consistently outperformed the other models in sensitivity, achieving 94.46% on 'PUPU' (see Figure 7). However, MobileNet-1D exhibited perfect specificity (100%) across all words, suggesting an absolute minimization of false positives, though at the cost of lower sensitivity (ranging from 71.43% to 80%). The CNN-1D baseline maintained a steady performance across both accuracy and sensitivity.

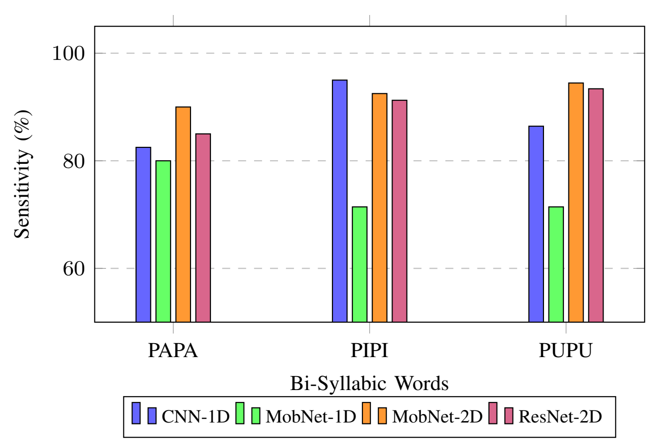

Fig. 7. Comparison of Model Sensitivity across Bi-syllabic Words.

TABLE I
CLASSIFICATION PERFORMANCE FOR SUSTAINED VOWELS (%)

Vowel	Classification Accuracy				Specificity				Sensitivity			
	CNN 1D	**MobNet 1D**	**MobNet 2D**	**ResNet 2D**	**CNN 1D**	**MobNet 1D**	**MobNet 2D**	**ResNet 2D**	**CNN 1D**	**MobNet 1D**	**MobNet 2D**	**ResNet 2D**
A	81.76	80.00	**90.20**	83.40	87.50	**90.00**	84.03	75.28	75.56	76.47	**96.25**	92.50
E	85.64	81.81	88.40	**89.00**	92.14	**100.00**	86.11	83.75	79.28	70.00	91.25	**93.75**
I	**81.21**	66.67	75.60	78.80	**80.22**	75.00	64.29	76.25	82.14	**84.62**	87.14	80.18
O	**88.33**	75.00	84.40	85.40	**89.28**	87.50	76.35	80.63	82.14	76.92	**92.50**	91.25
U	**84.68**	77.78	82.20	76.60	84.64	**85.40**	80.36	78.45	**85.48**	60.00	85.00	76.25

TABLE II
CLASSIFICATION PERFORMANCE FOR BI-SYLLABIC WORDS (%)

Word	Classification Accuracy				Specificity				Sensitivity			
	CNN 1D	**MobNet 1D**	**MobNet 2D**	**ResNet 2D**	**CNN 1D**	**MobNet 1D**	**MobNet 2D**	**ResNet 2D**	**CNN 1D**	**MobNet 1D**	**MobNet 2D**	**ResNet 2D**
PAPA	81.08	66.67	**87.20**	83.80	80.00	**100.00**	84.70	81.74	82.50	80.00	**90.00**	85.00
PIPI	**87.63**	88.89	86.20	86.60	77.64	**100.00**	77.17	80.00	**95.00**	71.43	92.50	91.25
PUPU	85.14	80.00	**87.20**	86.60	84.28	**100.00**	77.92	77.08	86.43	71.43	**94.46**	93.39

C. *Performance on Sentences*

Table III and Figure 8 outline the performance for continuous sentences (S0, S1). For these longer utterances, MobileNet-2D achieved high sensitivity (92.5%) and accuracy (82.8%) on S0. Interestingly, MobileNet-1D performed strongly on S1, achieving 100% specificity and 90.91% sensitivity. ResNet-2D struggled comparatively with sensitivity on continuous sentences, managing 65% and 70% on S0 and S1, respectively.

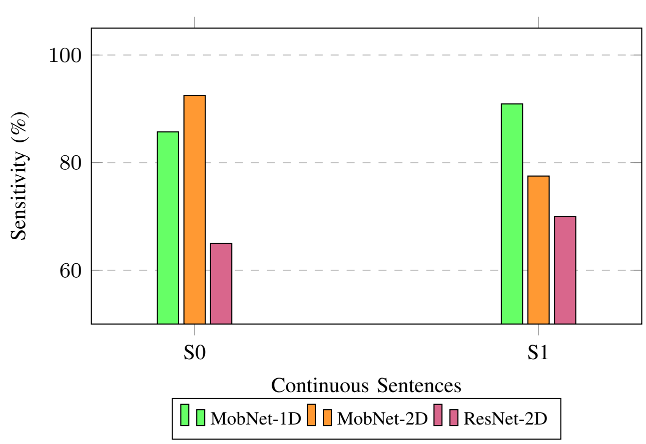

Fig. 8. Comparison of Model Sensitivity across Continuous Sentences.

TABLE III
CLASSIFICATION PERFORMANCE FOR SENTENCES (%)

Sent.	Accuracy			Specificity			Sensitivity		
	MN 1D	**MN 2D**	**RN 2D**	**MN 1D**	**MN 2D**	**RN 2D**	**MN 1D**	**MN 2D**	**RN 2D**
S0	80.00	**82.80**	75.00	80.00	73.22	**81.11**	85.71	**92.50**	65.00
S1	**80.00**	76.20	77.60	**100.00**	76.33	84.67	**90.91**	77.50	70.00

**MN = MobileNet, RN = ResNet*

VII. DISCUSSION

The conversion of audio streams into 2D spectrograms is extremely useful for recognizing specific speech problems associated with strokes. The 2D models (MobileNetV2 and ResNet-50) consistently outperformed the 1D baselines in *Sensitivity*, notably in sustained vowels and bi-syllables. This suggests that 2D models are unusually capable of reducing false negatives—an important criterion in medical diagnostic systems because failure to identify a stroke patient might delay crucial intervention.

The 1D-MobileNet design achieved flawless *Specificity* in word assessments, with no false positives. The decrease in specificity for the 2D models shows that background noise or artifacts generated during spectrogram augmentation (for example, flipping non-symmetrical audio features) may mislead the model into categorizing healthy variance as dysarthric.

Performance varies significantly with phonetic content. The formant patterns of sustained vowels, such as ”A” and ”E,” are quite stable and can be accurately represented by 2D spatial models. However, architectures that can depict sequential dependencies may benefit from the more intricate temporal transitions and dynamic articulatory patterns needed for whole sentences. This highlights the need for integrating static image-based feature representations with the temporal dynamical characteristics observed in continuous speech.

VIII. CONCLUSION

The effectiveness of spectrogram-based deep learning models for categorizing healthy and dysarthric speech at different phonetic complexity levels was investigated in this work. According to our experimental findings, the 2D convolutional architectures like MobileNet-2D and ResNet-2D consistently perform better than conventional 1D methods by capturing the rich time-frequency properties of speech signals. Strong feature learning is made possible by consistent spectral patterns in vowel and word-level tasks, where the performance gains are most evident. The suggested models maintain competitive performance and generalization capabilities, despite the fact that sentence-level classification is still challenging due to significant temporal unpredictability. Overall, the results show that lightweight 2D architectures successfully achieve a balance between accuracy and computational efficiency when applied to clinical speech analysis. Future studies might focus on using temporal modeling techniques to enhance performance in continuous speech scenarios.

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